

Revisiting Rational Broadcast Protocols

Shunya Otomo and Kenji Yasunaga

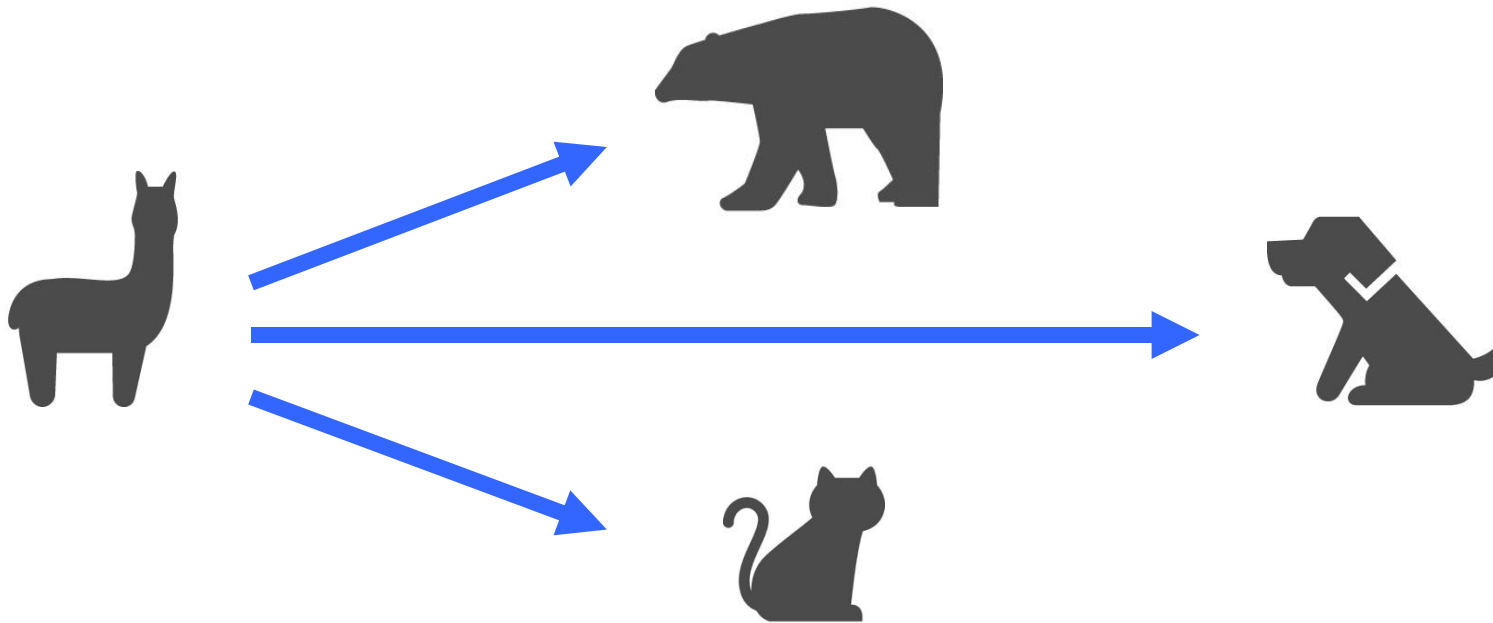


CANS 2025 @ Osaka, Japan

Broadcast Protocols

A sender delivers the **same message** to all parties even if the sender is **malicious**

- Fundamental building blocks for multiparty computation and blockchains



Broadcast Protocol (Setting & Requirements)

Setting

- A set $[n] = \{1, \dots, n\}$ of parties connected by secure P2P channels
- Adversary can corrupt $\leq t$ parties
- Synchronous communication (round-based communication)
- Authenticated setting: PKI (signature scheme) is available

Requirements

- **Validity:** If a sender $s \in [n]$ with input m is honest, then all honest parties output m
- **Agreement:** All honest parties output the same value

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- **Randomized** protocols for $t \geq n/2$ (honest minority)
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Wan et al. (2020) for $t \leq (1 - \epsilon)n$ for any $\epsilon > 0$

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Wan et al. (2020) for $t \leq (1 - \epsilon)n$ for any $\epsilon > 0$
- Lower bounds for **randomized** protocols (Garay et al., 2007)
 - No $O(1)$ -round protocol for $t = (1 - o(1))n$

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Yamashita-Yasunaga (2023)

- $O(1)$ -round **deterministic** protocol for any $t < n$ against **rational** adversaries

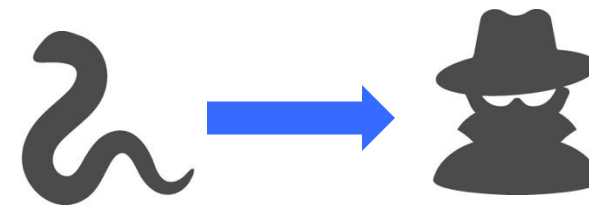
Rational adversary tries to **maximize its utility**



Detection-averse adversary prefers to attack the protocol only when it can avoid being detected (i.e., avoids actions that would trigger detection)

This Work

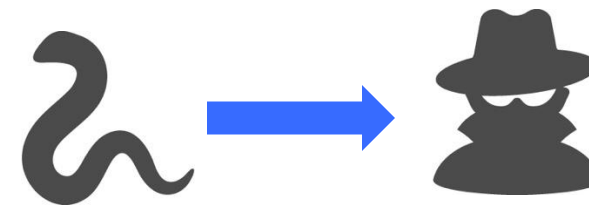
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We revisit the YY23 protocol

- Observation: YY23 is secure against **general rational** adversaries
 - It is secure even without relying on any attack-detection mechanism

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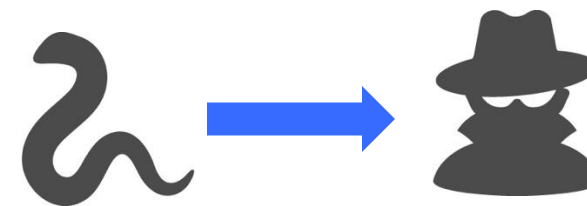
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We construct:

Protocol Π_1

Protocol Π_2

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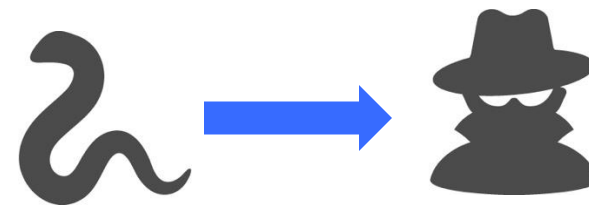


- **$O(1)$** -round protocol with **enhanced security** for **rational** adversaries
 - For **$t \geq n/2$** , YY23 achieves only **weak validity** and **may fail attack detection**
- Π_1 is a **$(t+4)$** -round protocol against **malicious** adversaries

Protocol Π_2



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Protocol Π_2

- **$O(1)$** rounds for **detection-averse** adversaries
- **Optimal** round-complexity **$t+1$** for **malicious** adversaries



Comparison with YY23

Protocol	Resiliency	Attack Detection	# Rounds	Adversary Model	Security Requirements
YY23	$t < n/2$	✓	4	Rational	Validity & Agreement
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			$t + 1$	Malicious	Validity & Weak Agreement

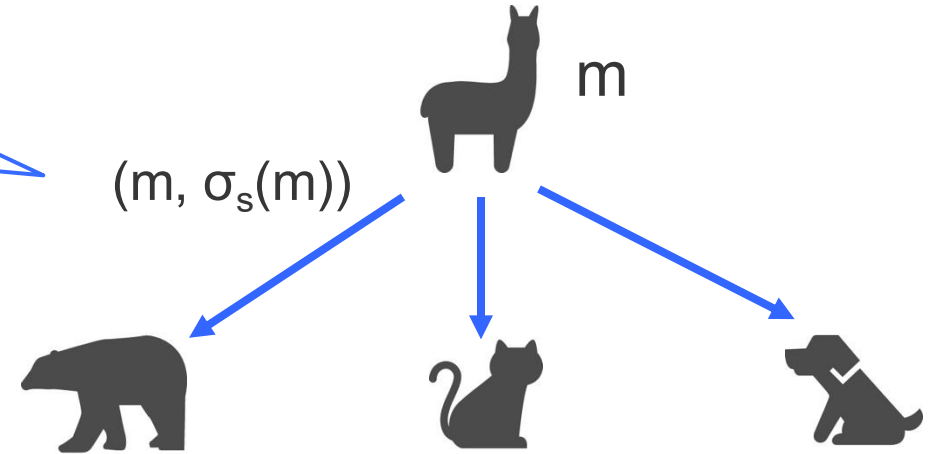
YY23 Protocol



YY23 Protocol

Round 1:

Sender $s \in [n]$ sends $(m, \sigma_s(m))$ to all parties



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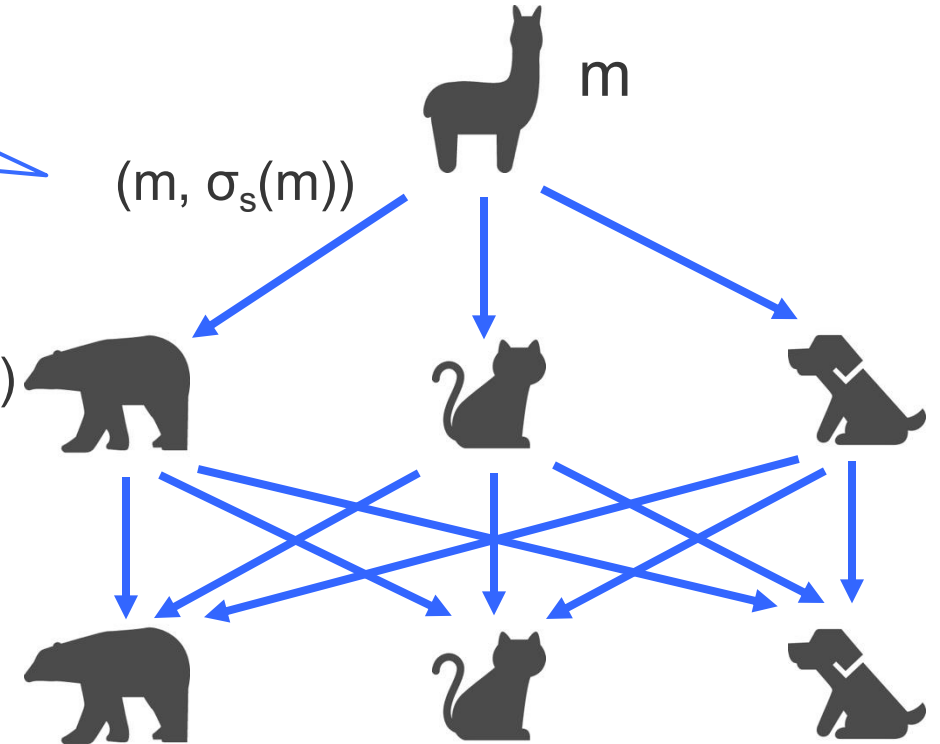
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Countersignature $(m, \sigma_i(\sigma_s(m)))$

Round 2:

Each $i \in [n]$ sends a countersignature $(m, \sigma_i(\sigma_s(m)))$ to all parties



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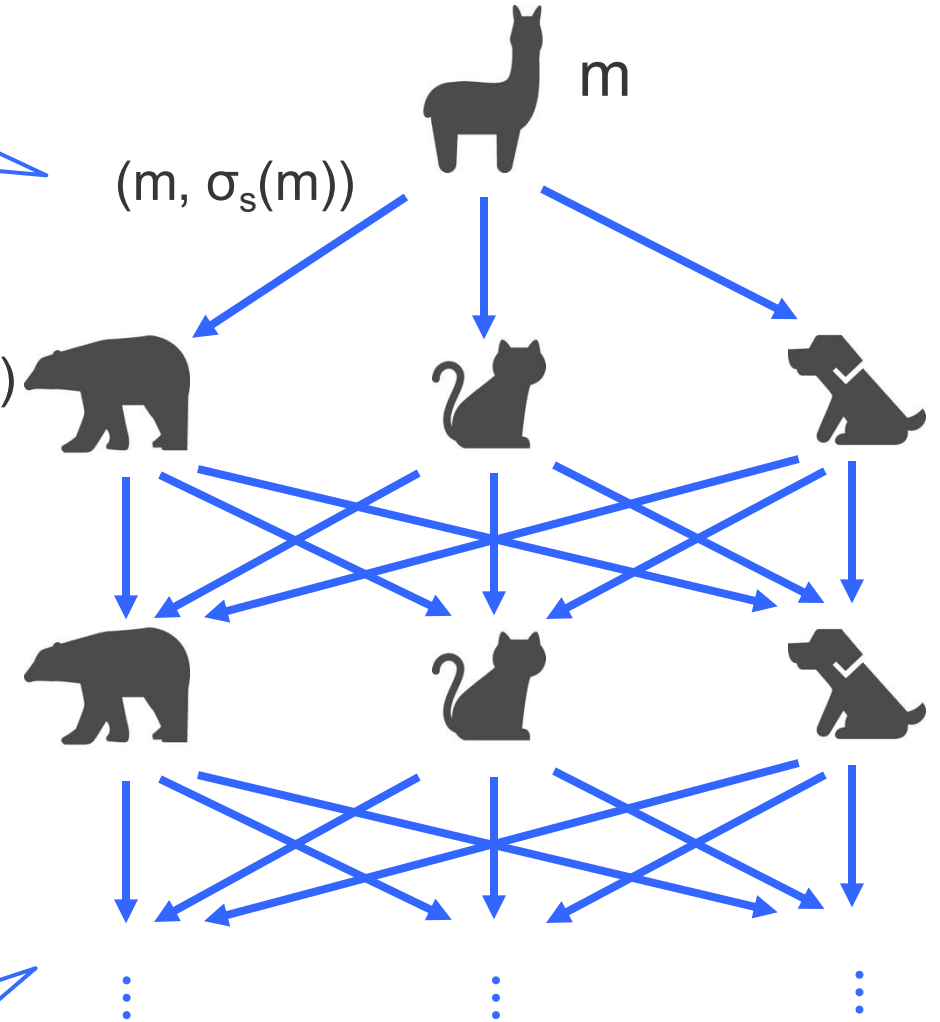
PoD_m^i

$t+1$

$(m, \sigma_B(\sigma_s(m)))$
 $\vdots$
 $(m, \sigma_D(\sigma_s(m)))$

Round 3:

Each $i \in [n]$ collects $t+1$ valid countersignatures to generate a **proof of dissemination** PoD_m^i , and sends it to all parties



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Round 2:

Each $i \in [n]$ sends a **countersignature** $(m, \sigma_i(\sigma_s(m)))$ to all parties

At least one honest countersignature was disseminated to everyone

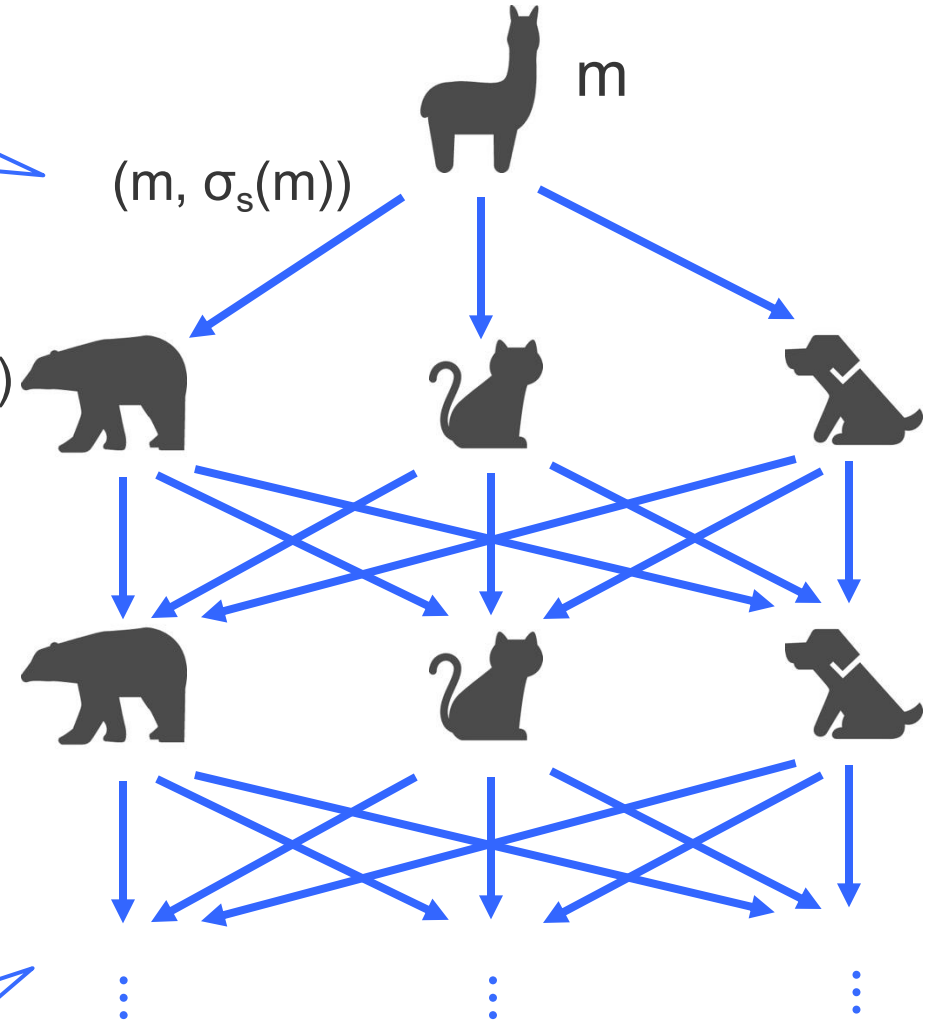
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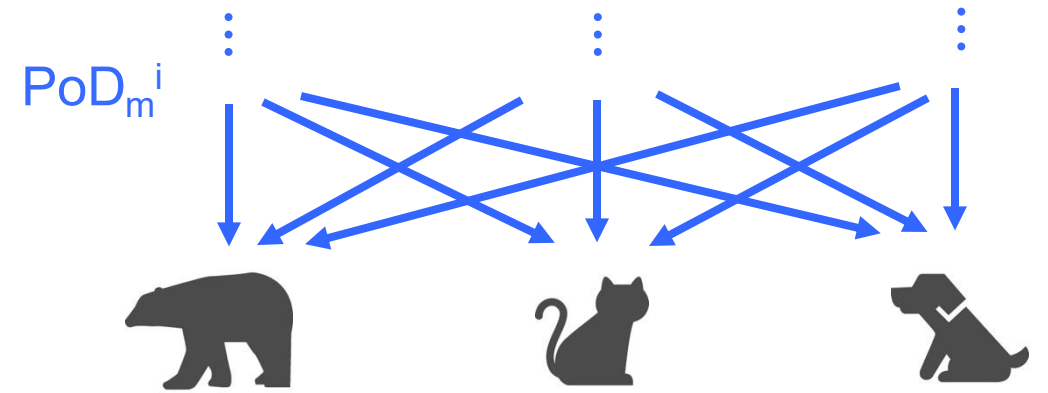
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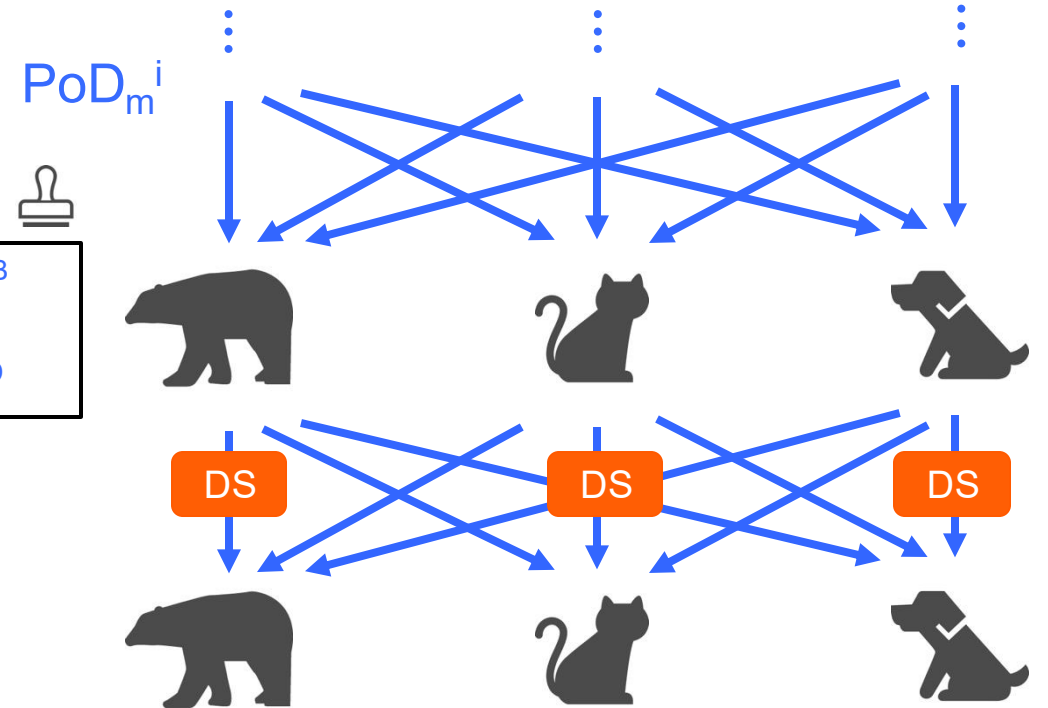
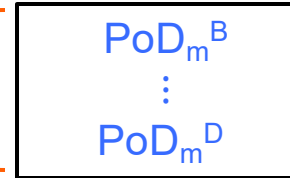
YY23 Protocol

Round 4:

Each $i \in [n]$ collects $t+1$ valid PoD_m^j to generate a *proof of agreement* PoA_m^i , and sends it using the Dolev-Strong (DS) protocol

PoA_m^i

$t+1$



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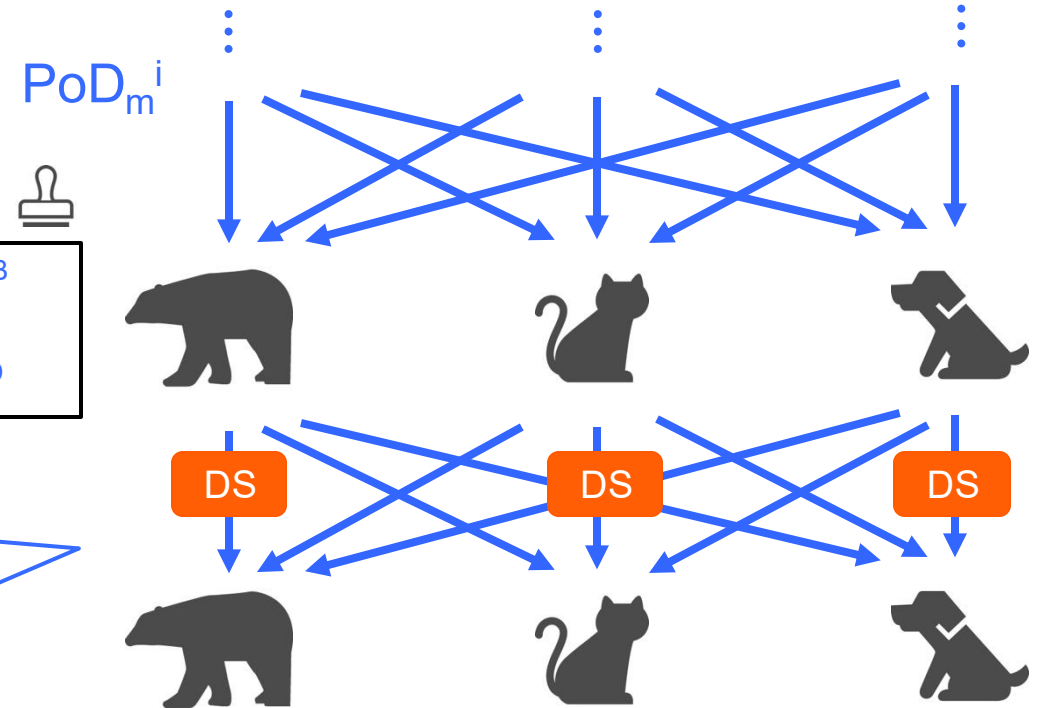
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If $i \in [n]$ collects $t+1$ valid PoA_m^j , it outputs m and halts. Otherwise, i continues the DS protocol



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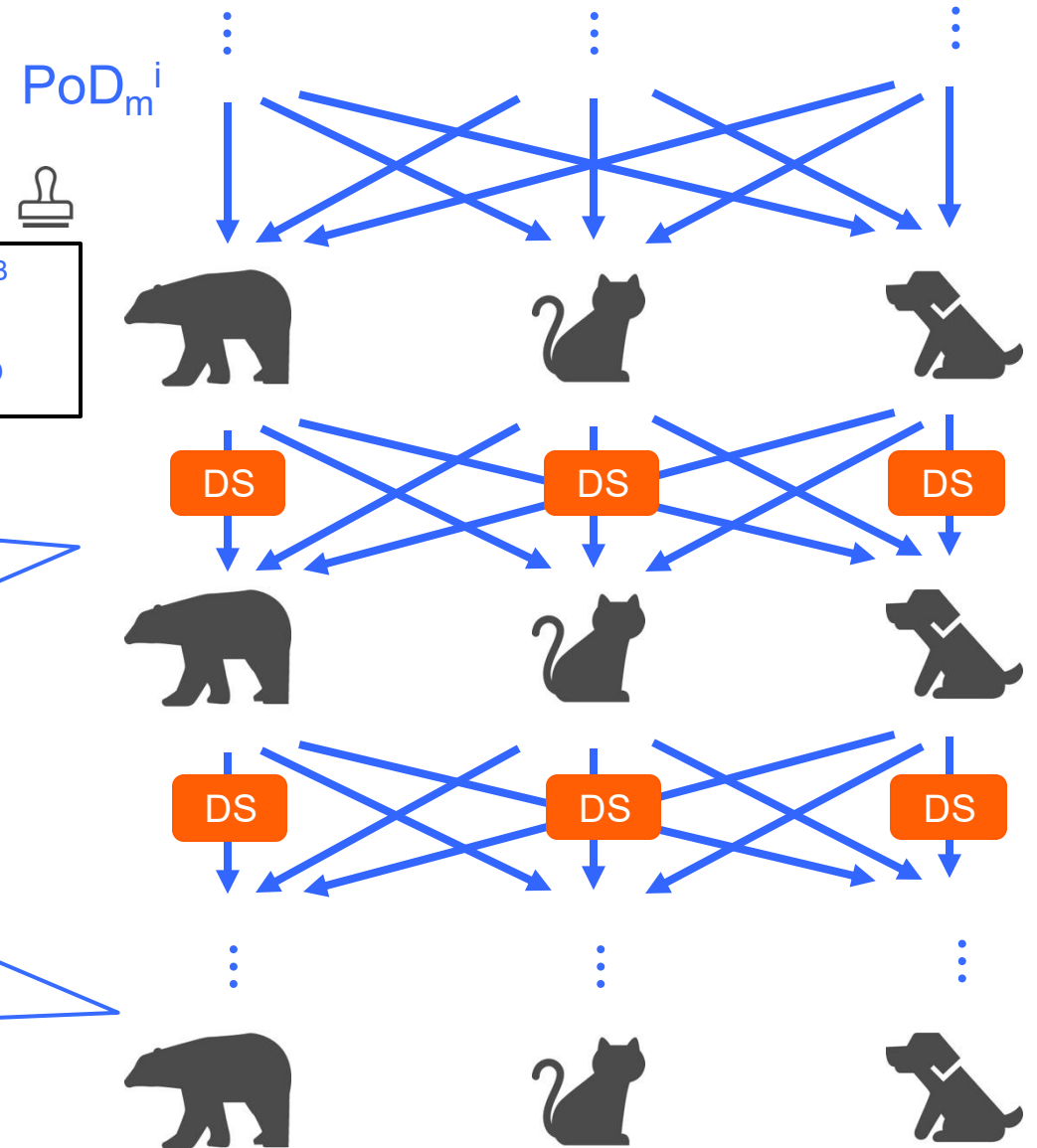
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End of Round 4:

If $i \in [n]$ collects $t+1$ valid PoA_m^j , it outputs m and halts. Otherwise, i continues the DS protocol

End of Round $t+4$:

If the DS protocol outputs a valid PoA_m , i outputs m . Otherwise, i outputs $\perp$ and regards s as cheating



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- A rational adversary has no incentive to attack the protocol, since it cannot improve its utility by violating the security requirements
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Interpretation:

- The $O(1)$ -round termination comes from the inherent security of the YY23 protocol, not from the attack-detection mechanism

Two Weaknesses of the YY23 Protocol

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For $t \geq n/2$, an honest sender s can be falsely detected as a cheater

How?

- If $t = n/2$ parties simply do nothing, then valid PoDs cannot be generated
- Honest parties fail to receive valid PoDs (or PoAs) and output $\perp$
→ s is incorrectly regarded as a cheater

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- If **s** is honest, honest parties never output **⊥**

- No false detection

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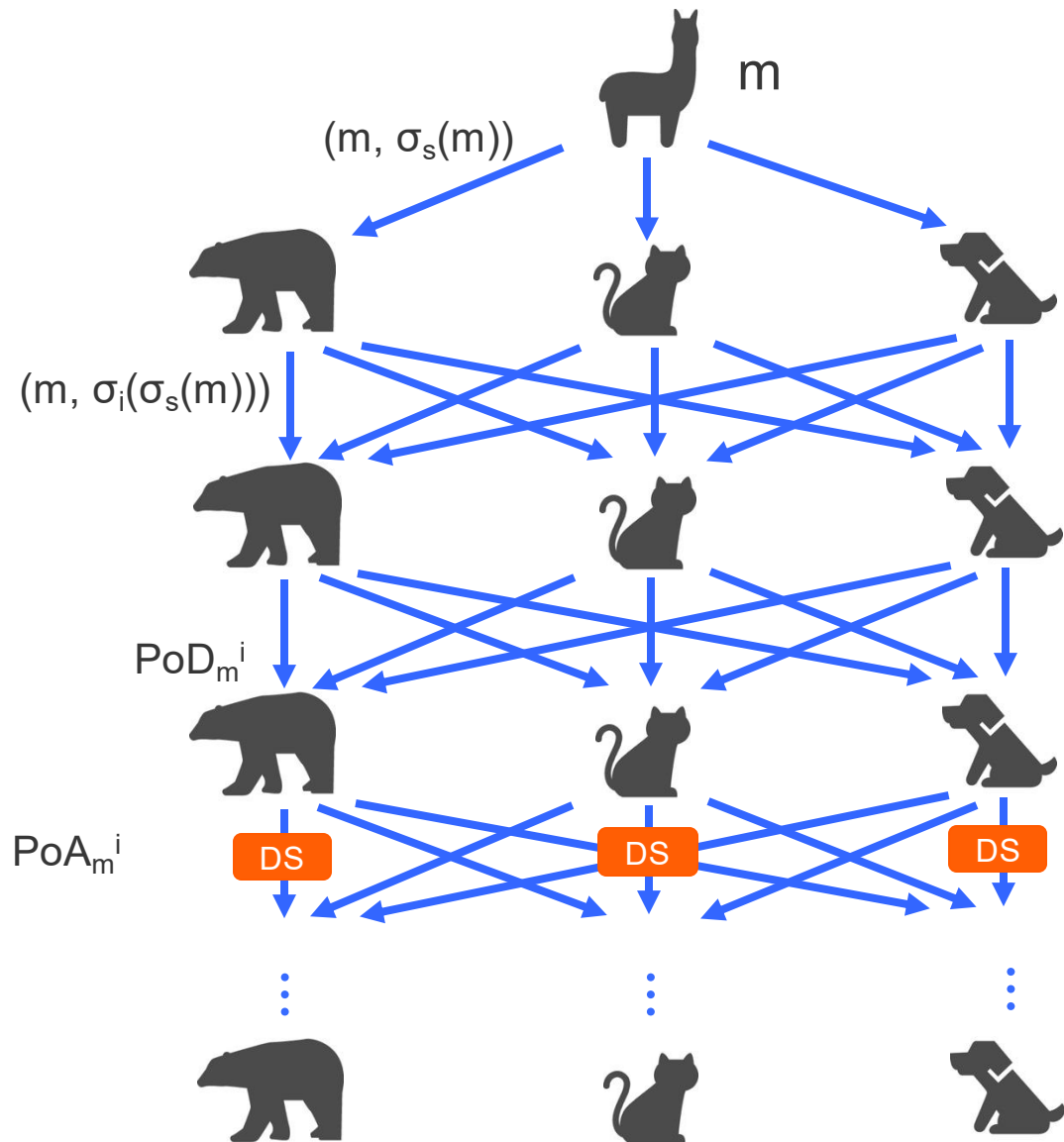
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How to achieve standard validity?

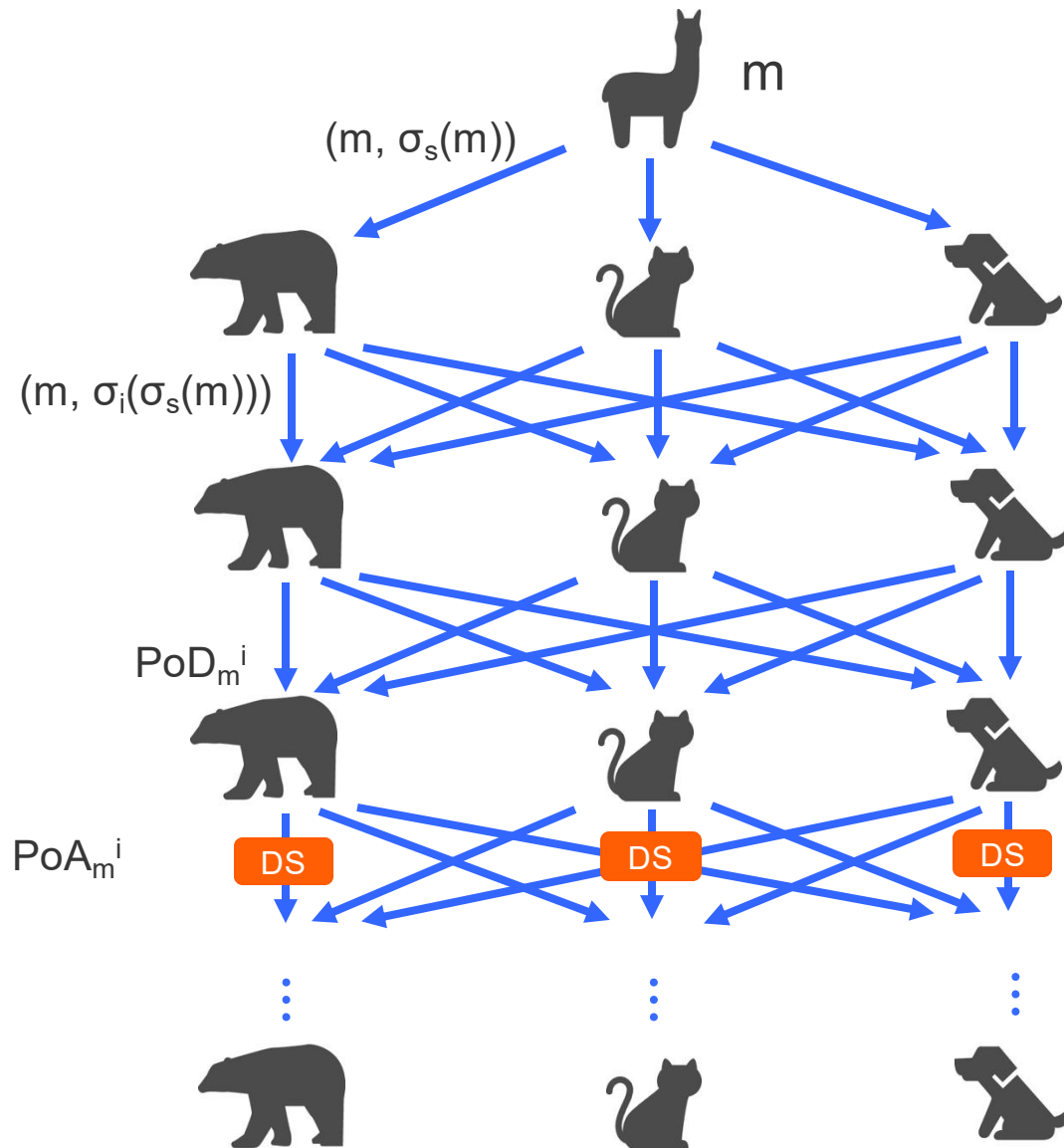
Protocol Π_1

YY23 protocol

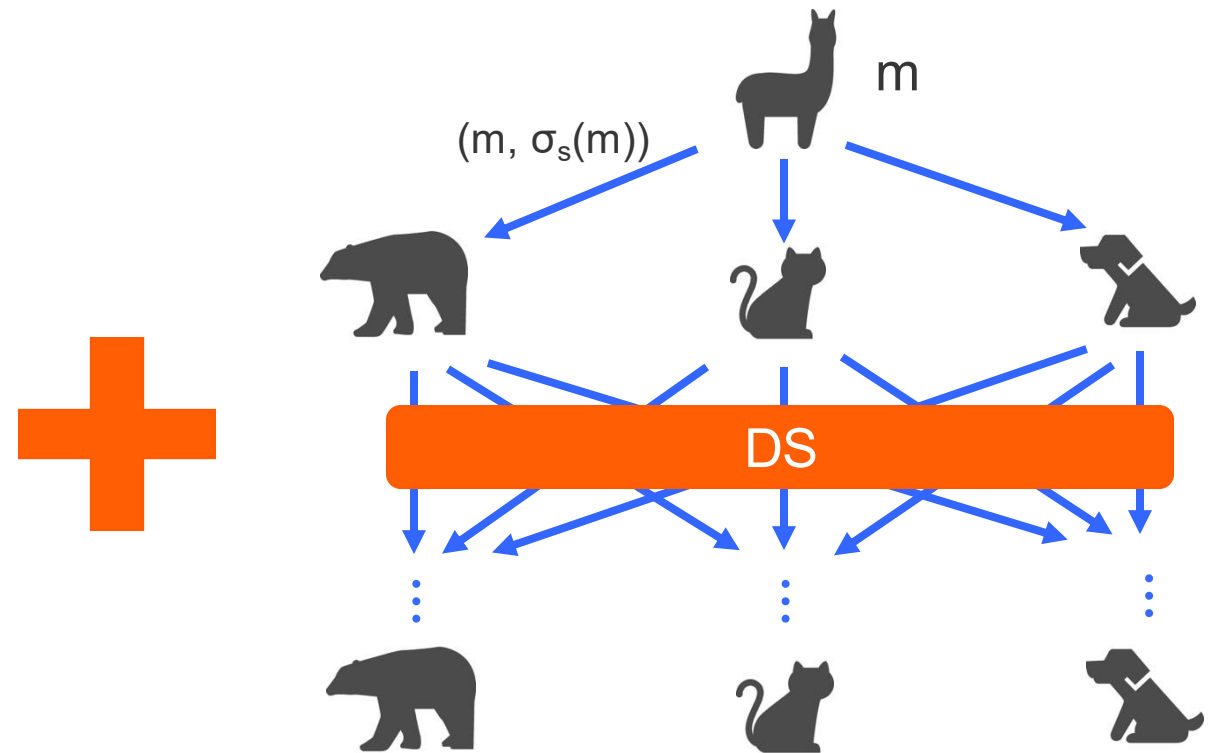
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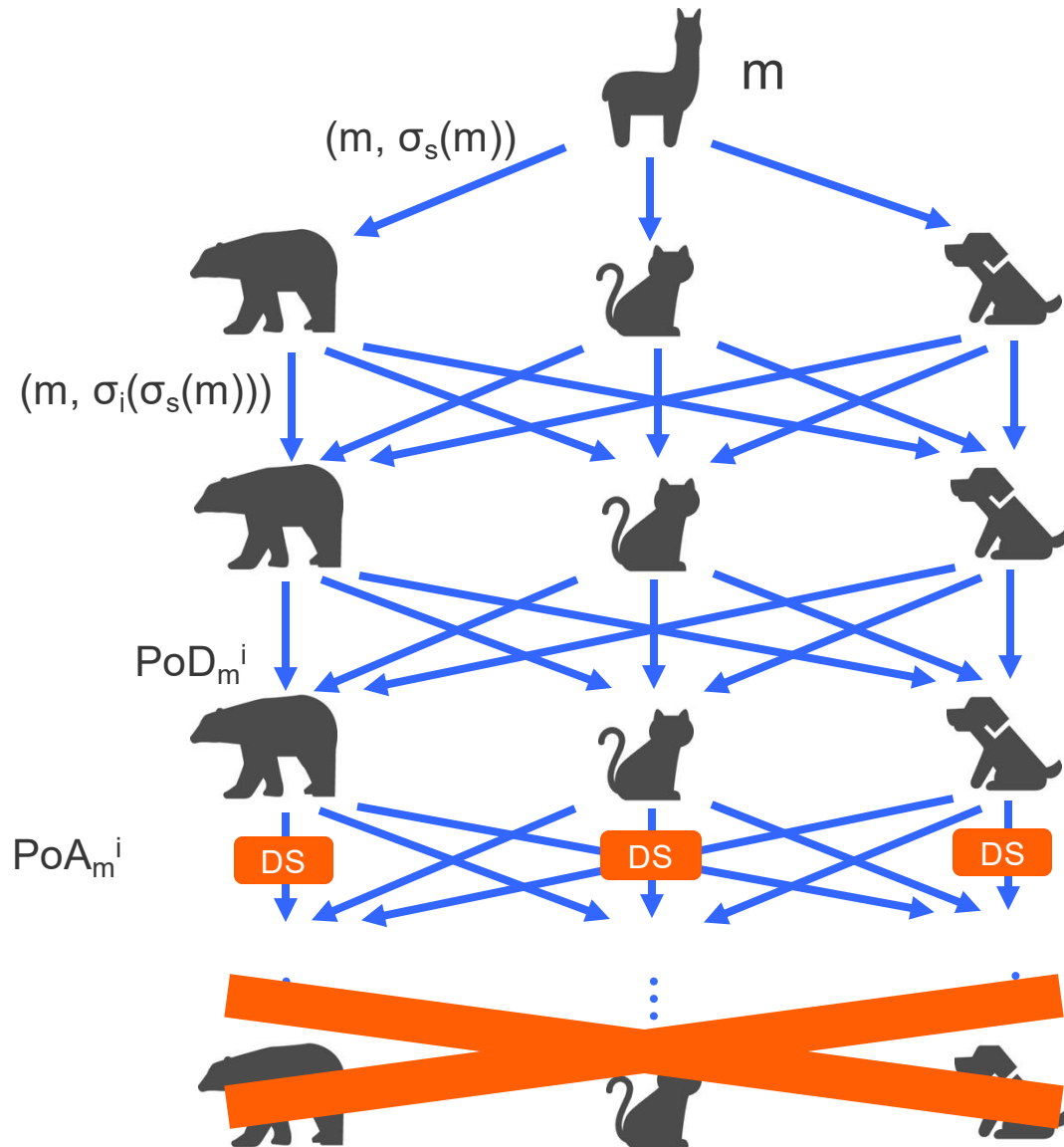
Dolev-Strong (DS) protocol



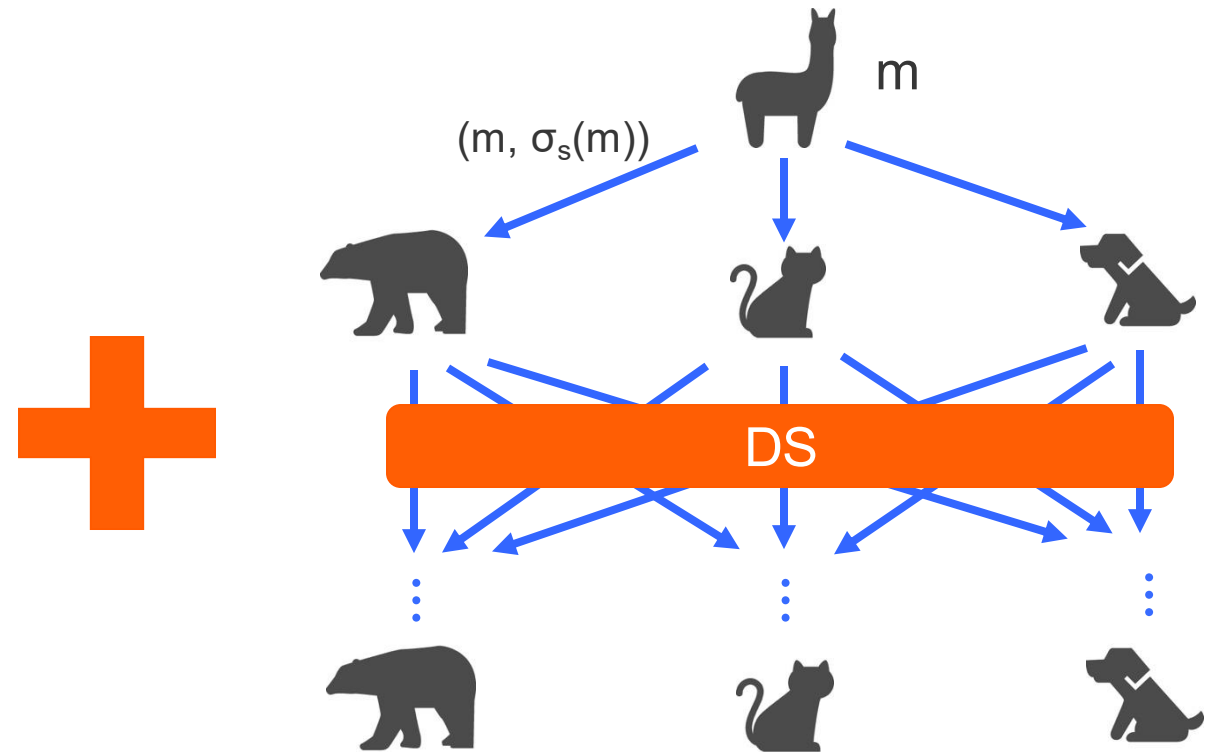
- Run the DS protocol *in parallel* with the YY23 protocol
- If the YY23 output is $\perp$
 → Use the DS output instead

Protocol Π_2

YY23 protocol

Protocol Π_2

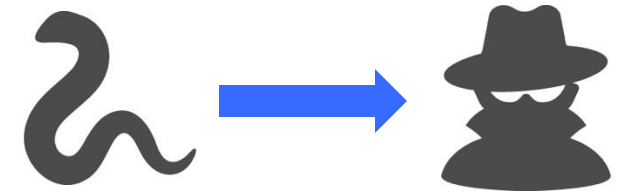
Dolev-Strong (DS) protocol



- Stop the YY23 protocol after Round 4
- **Detection-averse** adversary avoids **weak agreement** ($out_i = m, out_j = \perp$) since such disagreement triggers detection

Summary

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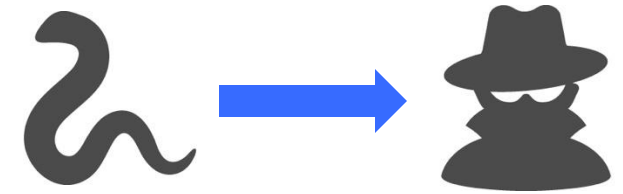
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